CAPACITOR AND INDUCTOR TRANSIENT CURVE

AC CIRCUITS

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GRADING SYSTEM

FINAL GRADE

30%	30%	**40%**
PRELIM	MIDTERM	ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Sinusoidal Waveform

Three Basic Types of Circuits

- R-Circuit
- C-Circuit
- L-Circuit

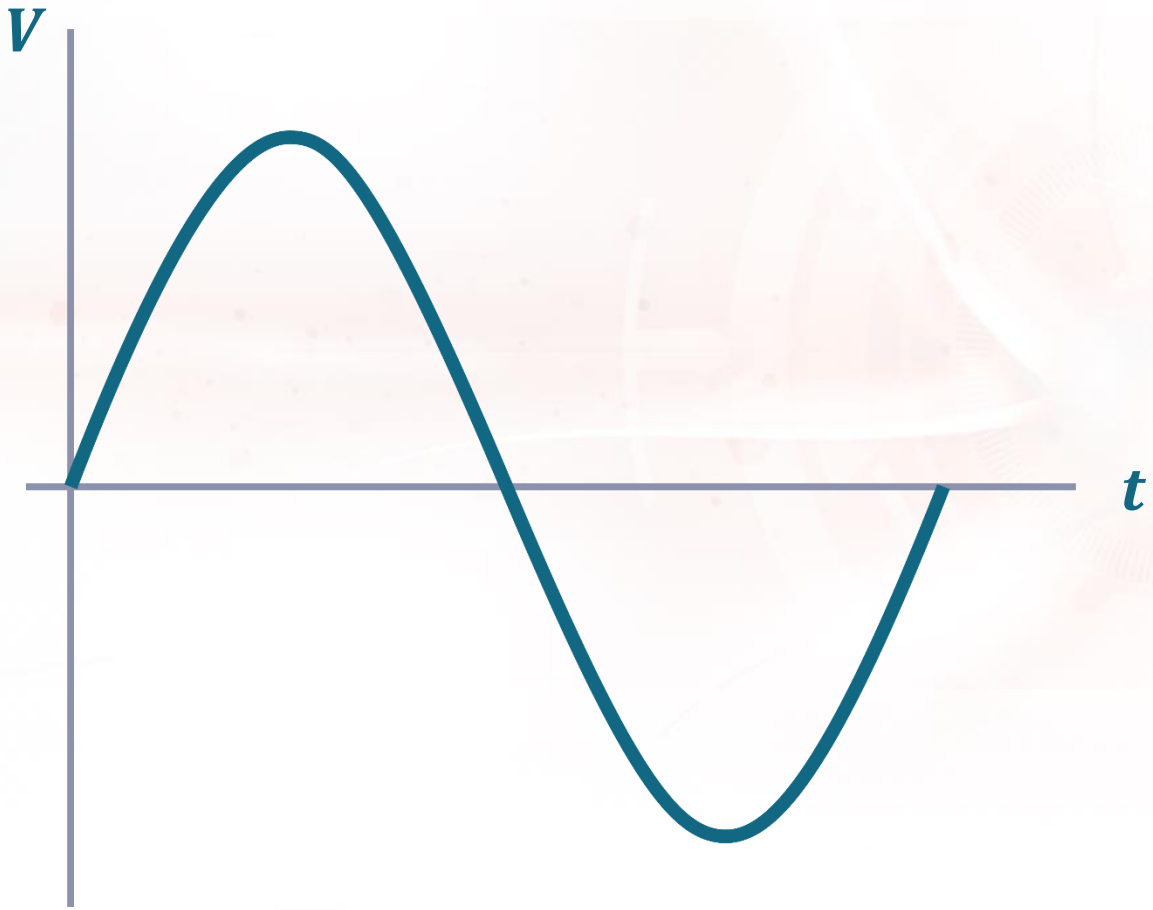
Capacitor Transient Curve

Inductor Transient Curve

SINUSOIDAL WAVEFORM

INSTANTANEOUS VOLTAGE

$$v(t) = V_p \sin \omega t$$



where,

$v(t) = instantaneous\ voltage, V$

$V_p = peak/maximum\ voltage, V$

$\omega = 2\pi f = angular\ speed, rad/s$

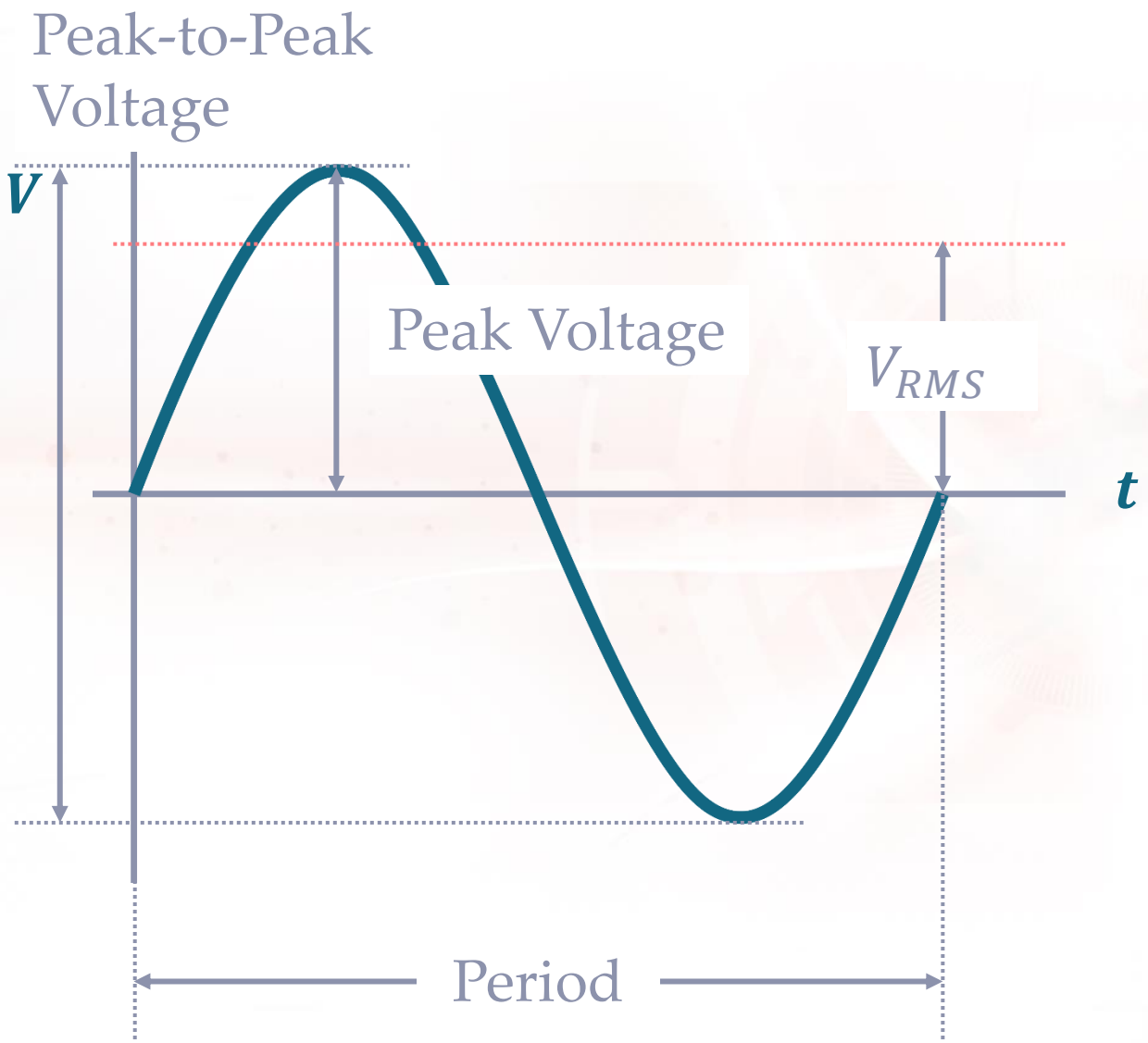
$f = frequency, Hz$

$t = time, s$

EXERCISE

A **60-cycle** current has a maximum value of **6.5A**. What will be the instantaneous value of **current** **2.5ms** after the wave passes through zero in a **positive** direction?

SINUSOIDAL WAVE VALUES



Root-Mean-Square,

$$V_{RMS} = \frac{V_p}{\sqrt{2}}$$

Period,

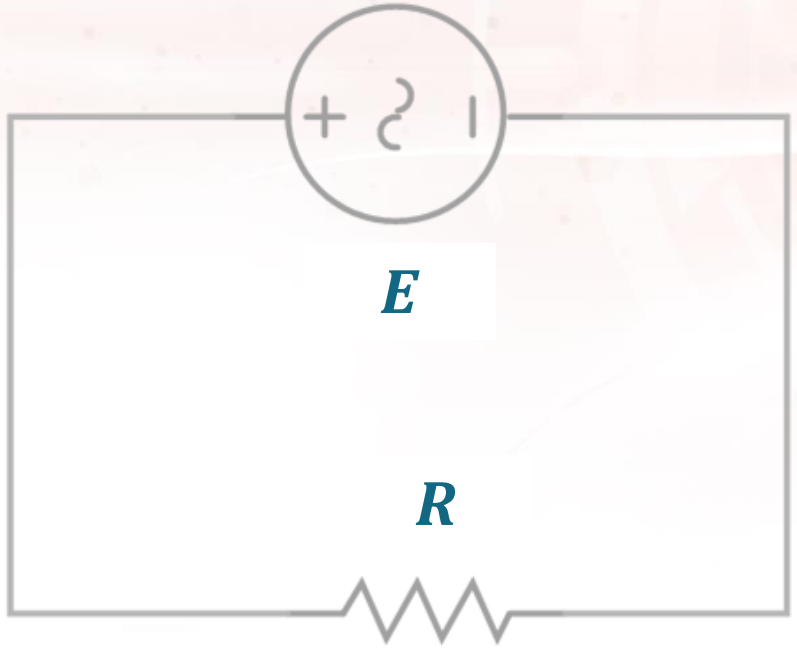
$$T = \frac{1}{f}$$

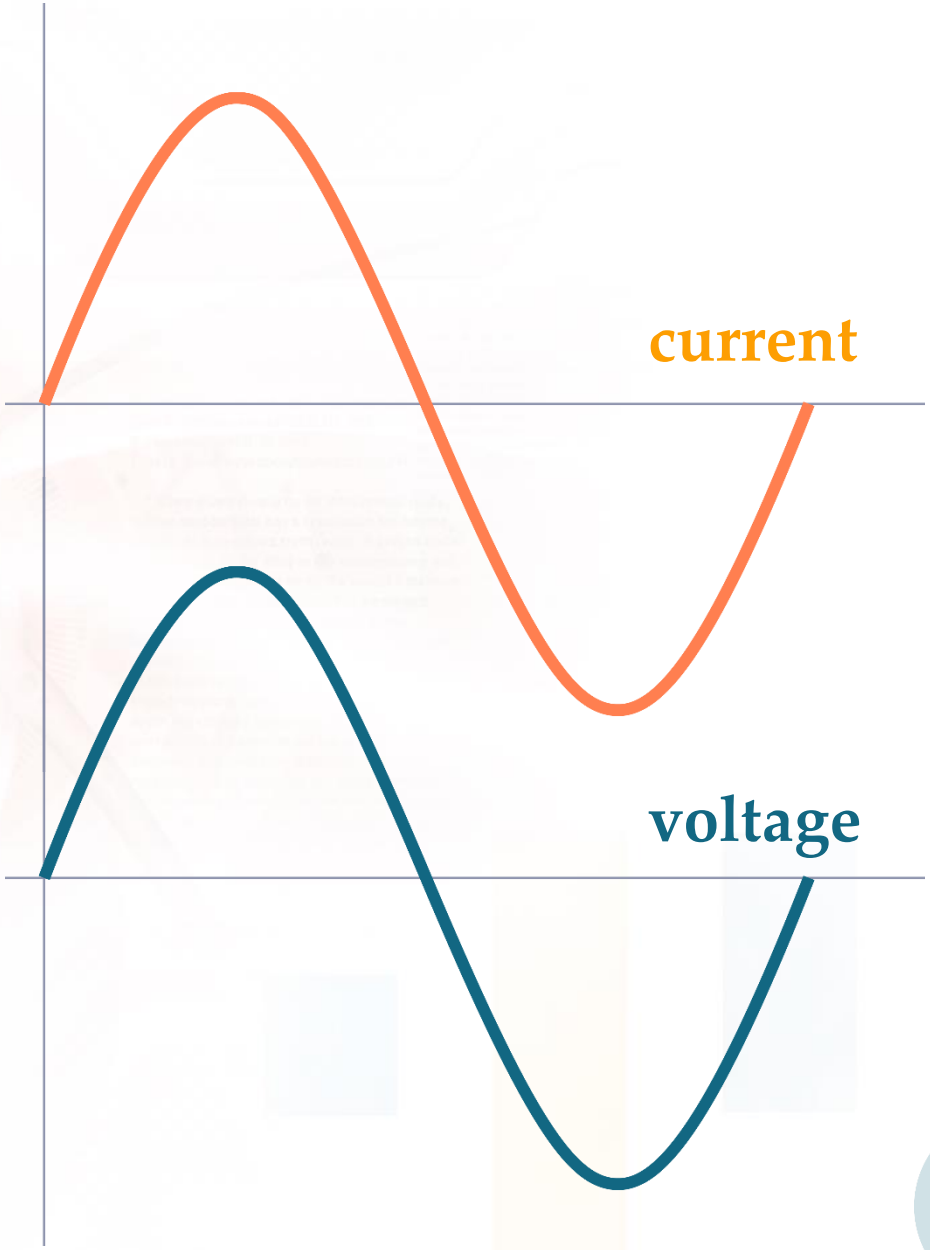
EXERCISE

The **rms** value of the voltage in a **60-cycle** circuit is **115 volts**. Write the **equation** for the sinusoidal wave.

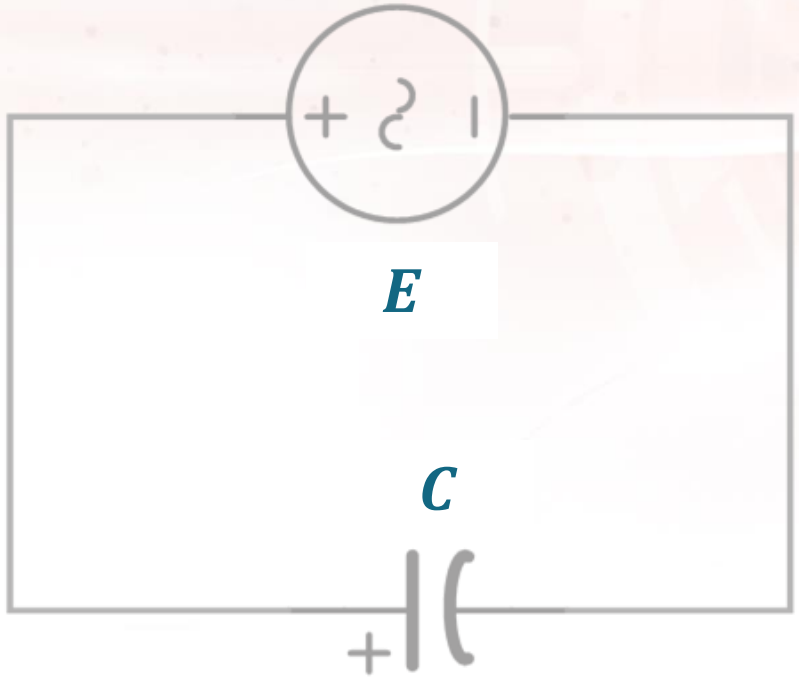
THREE BASIC TYPES OF CIRCUITS

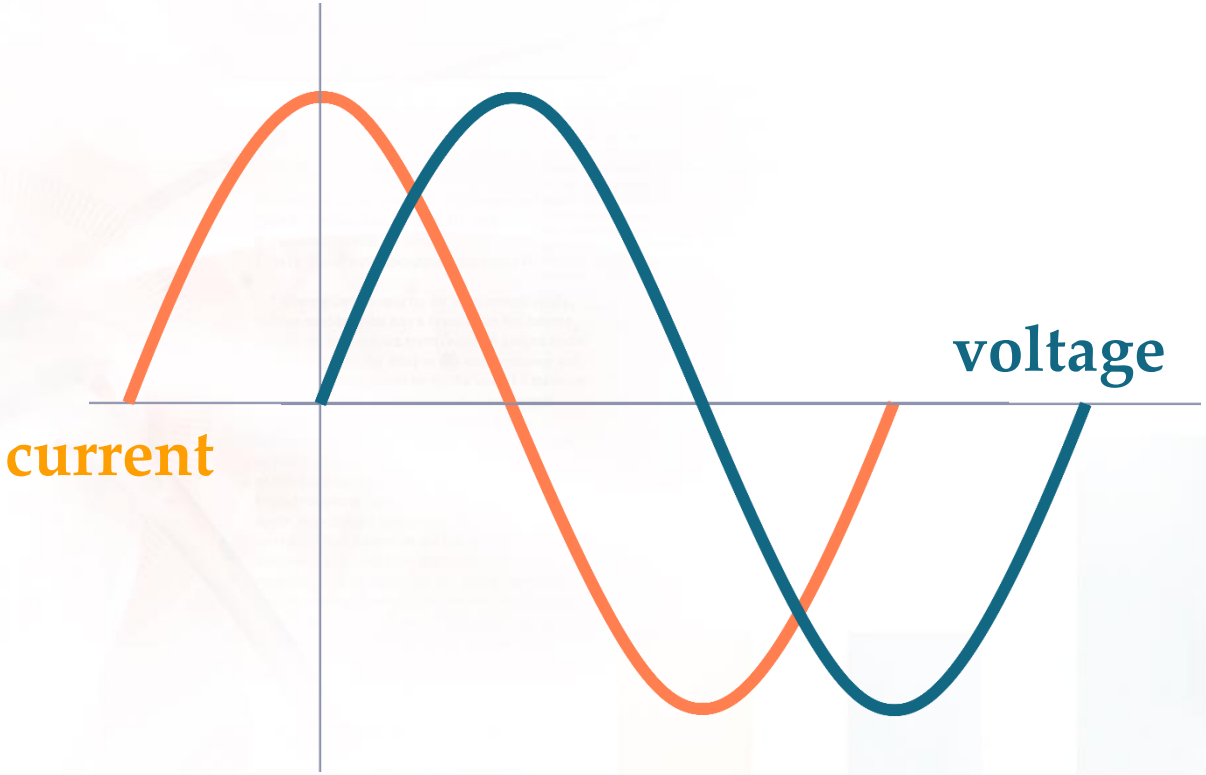
The behavior of a pure resistor in an AC circuit is exactly similar to that in a DC circuit. Current and voltage phasors are always **in phase** in a resistor circuits.



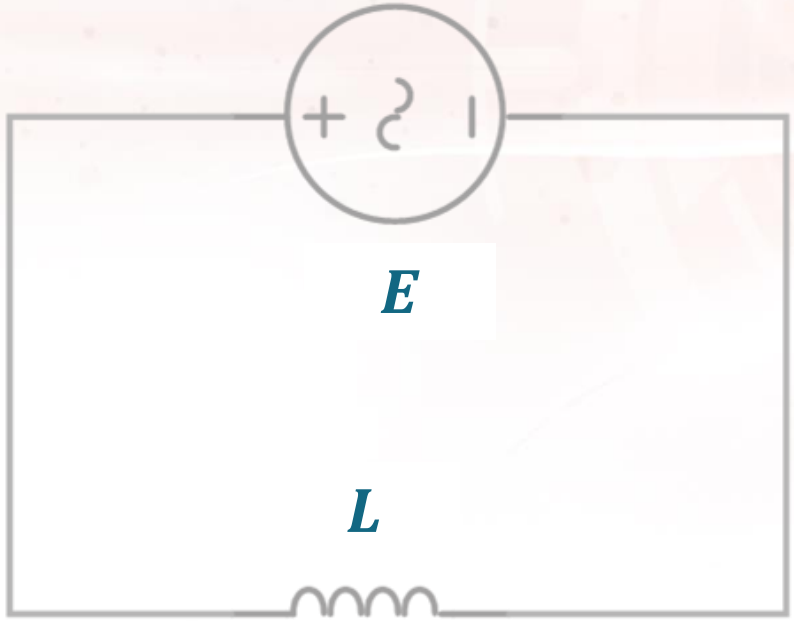


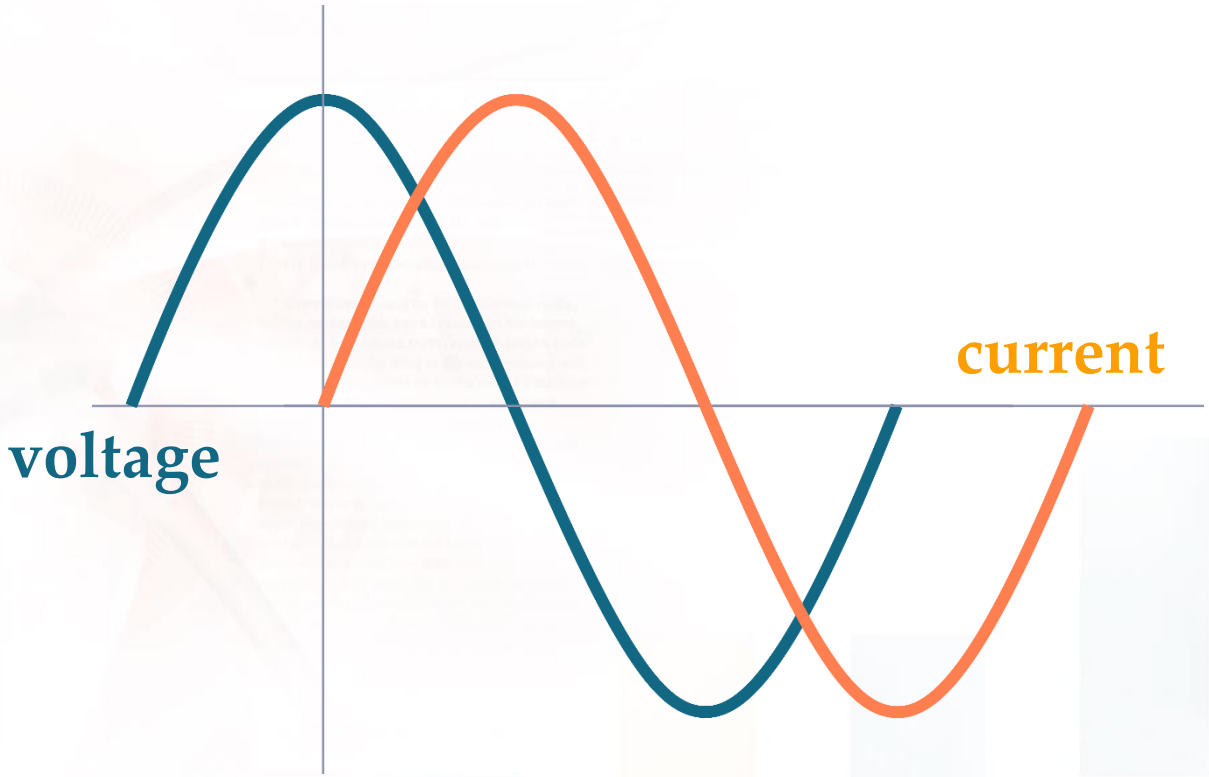
A perfect capacitor in an AC circuit takes a **current** that **leads** the impressed emf by exactly 90 elec deg.





A pure inductor in an AC circuit takes a **current** that **lags** behind the impressed emf by exactly 90 elec deg.



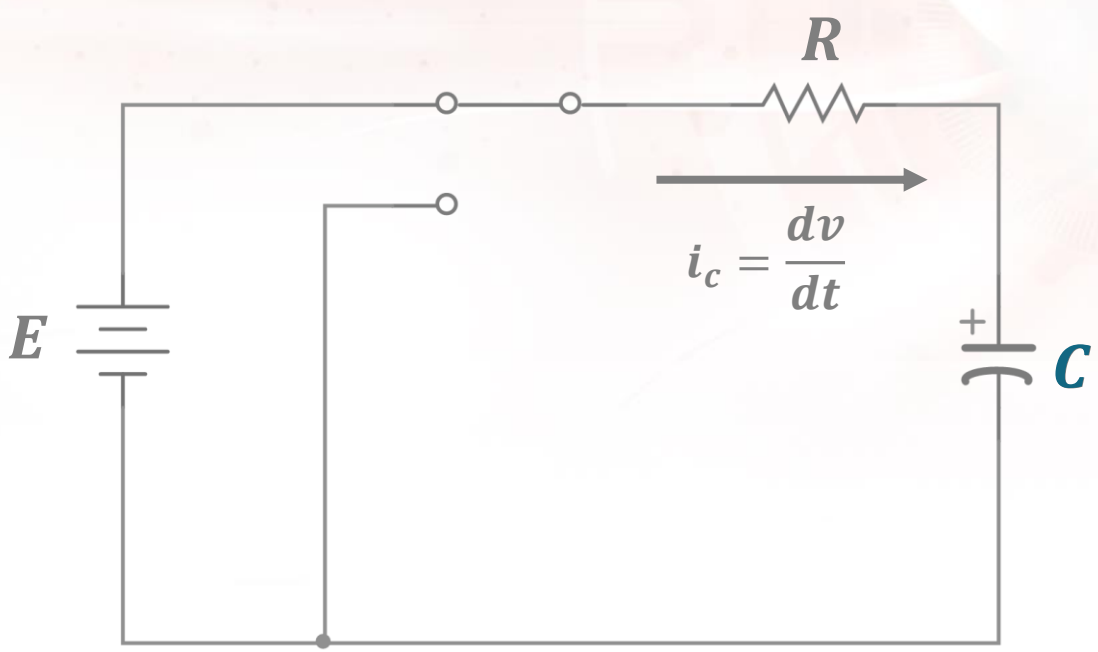


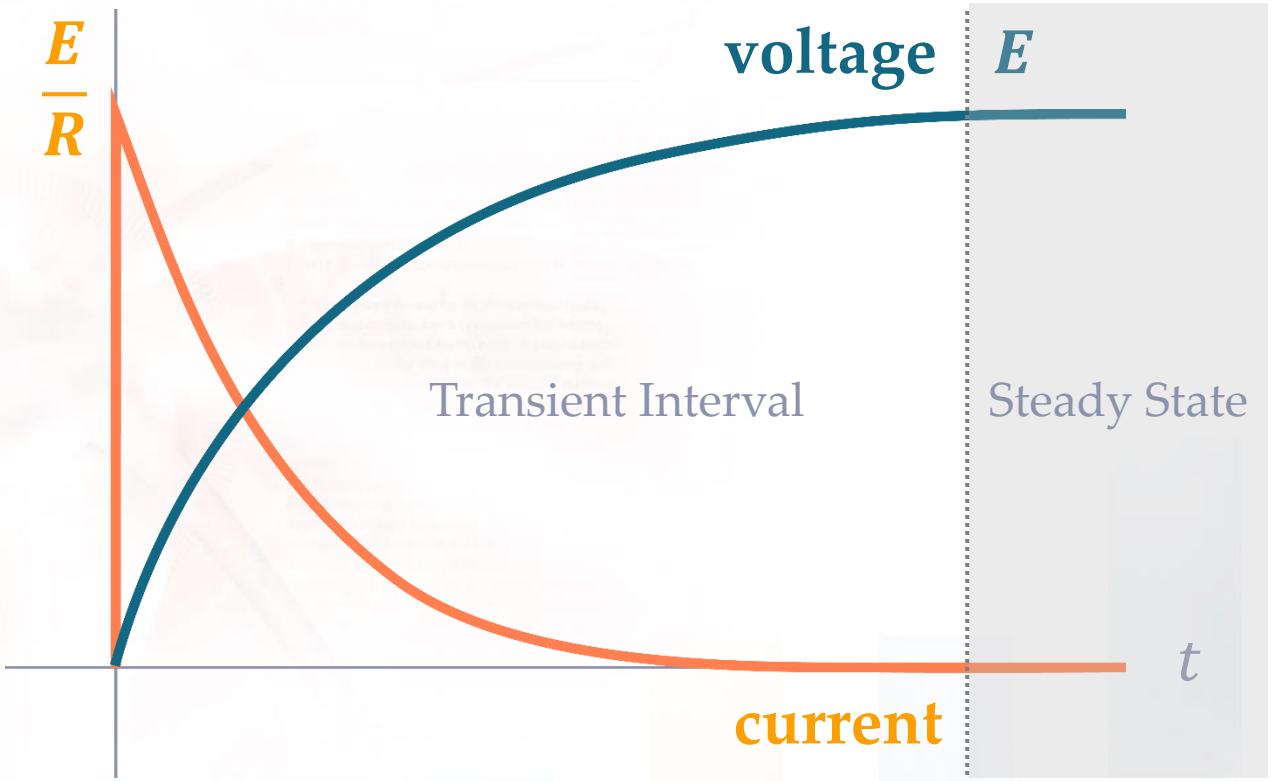
CAPACITOR

TRANSIENT CURVE

Capacitor **current** instantly jumps to $\frac{E}{R}$ and exponentially decays to **zero**

Capacitor **voltage** begins at **zero** and exponentially increases to E **volts**

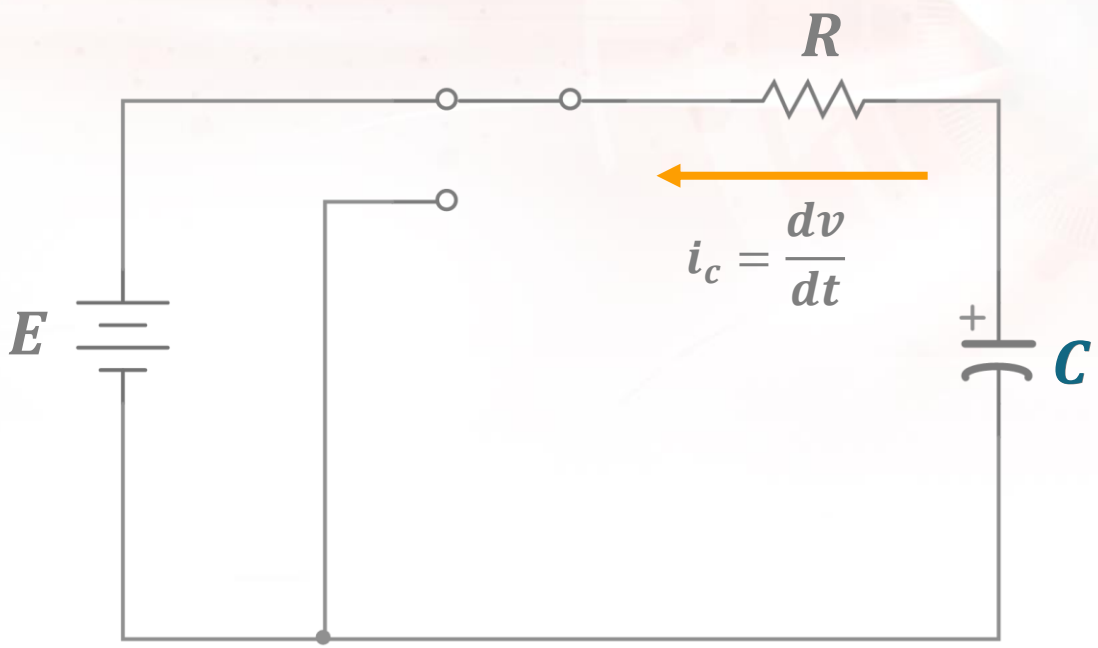


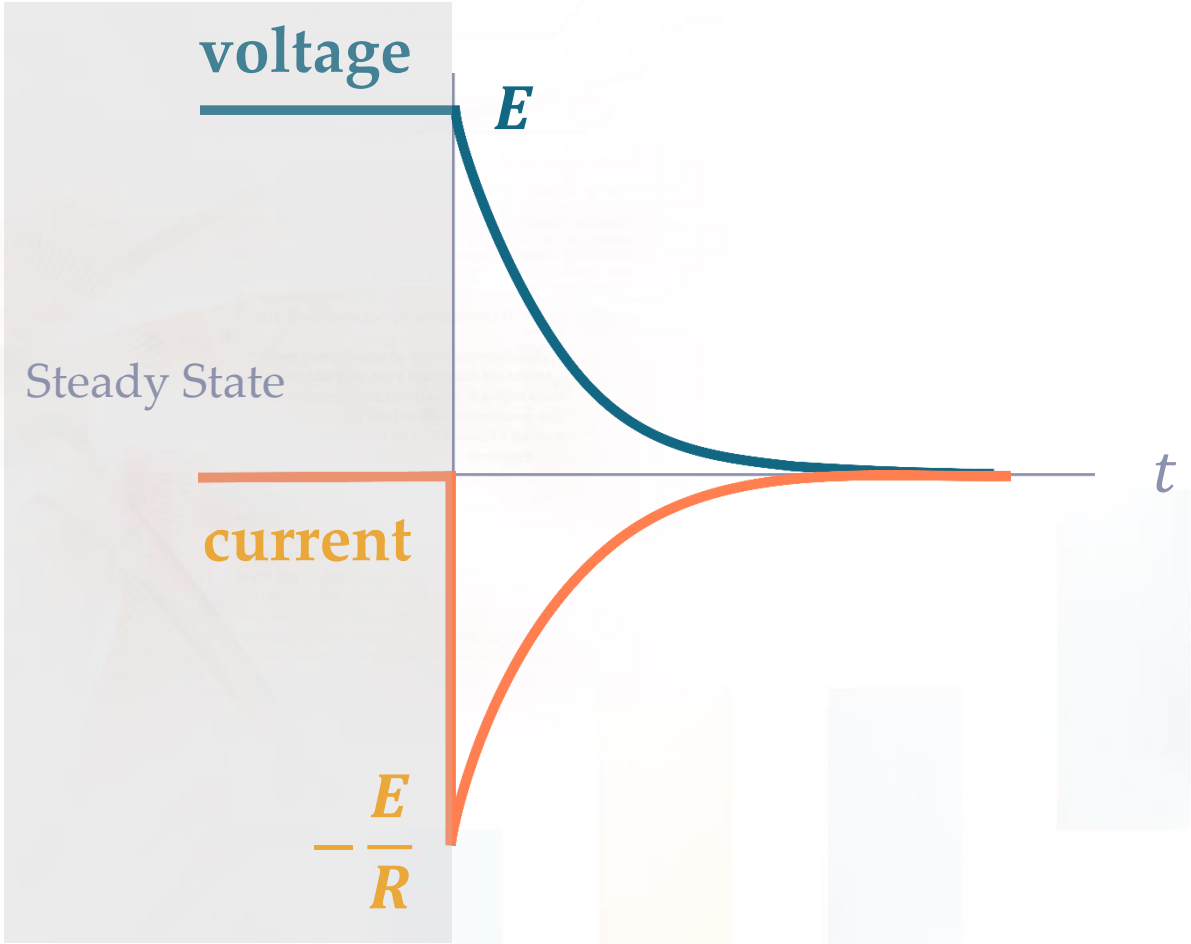


Capacitor **voltage** has E Volts

Capacitor **current** instantly jumps to $-\frac{E}{R}$

Both voltage and current will decay exponentially to **zero**

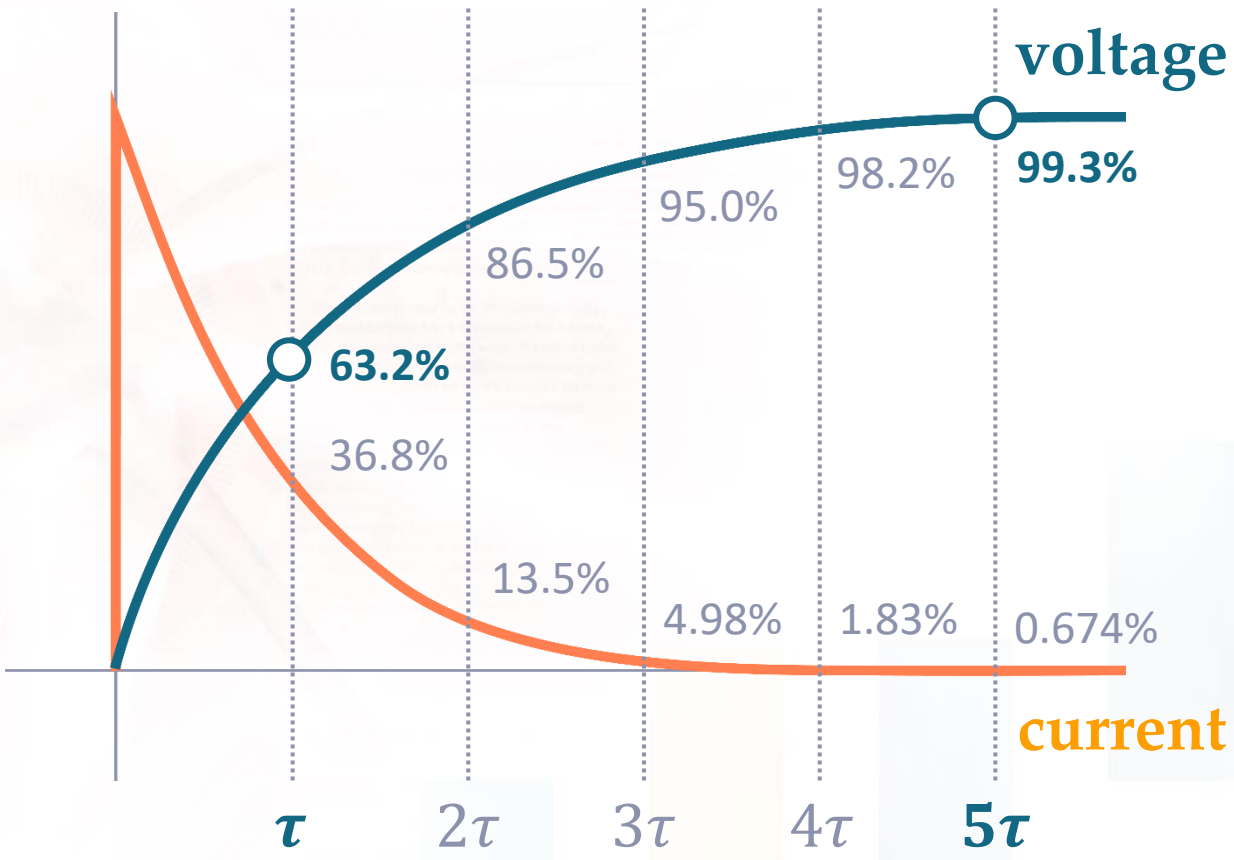




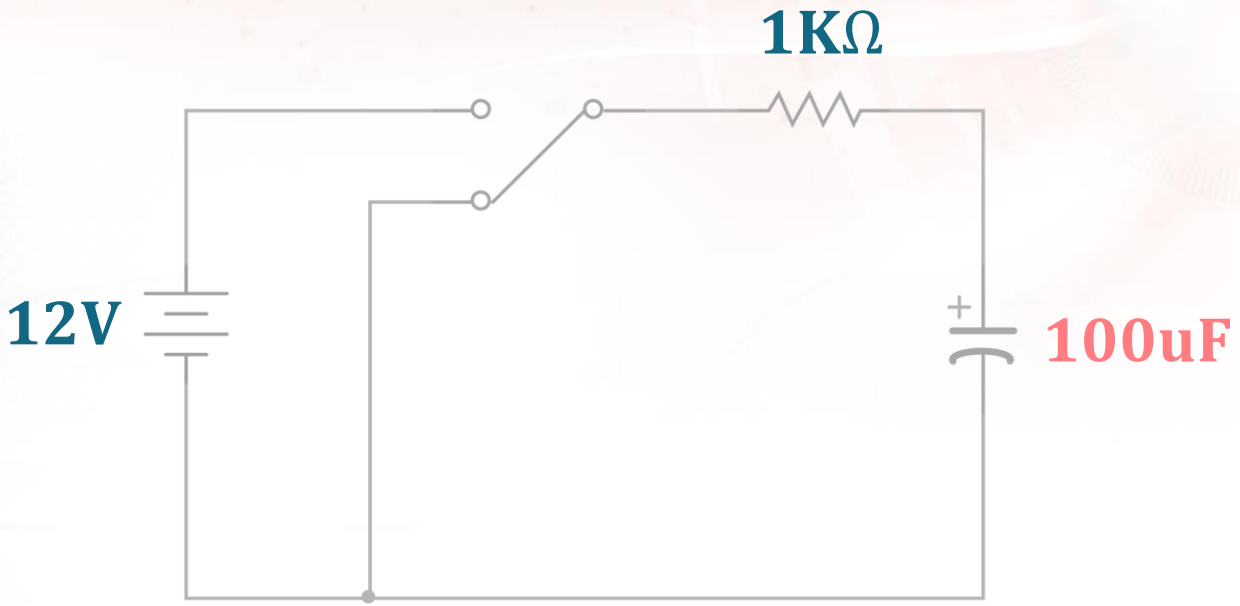
The rate at which capacitor charges depends on the product of R and C

$$\tau = RC$$

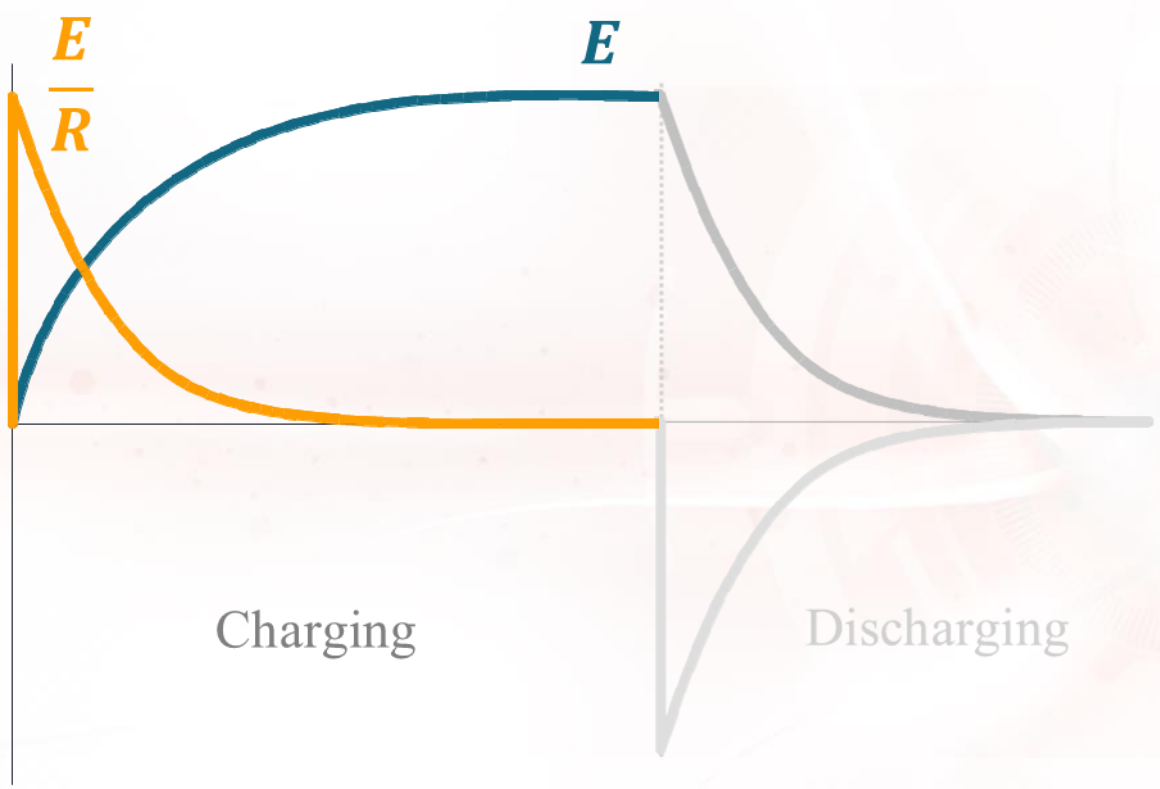
Transient can be considered to last for only five time constants, $\mathbf{5\tau}$.



A **100uF** capacitor is connected to a **12V** DC power supply through a resistor of **1KΩ**. Determine the **time** it takes for the capacitor to charge to **86.5%** of its maximum voltage.



CAPACITOR CHARGING EQUATIONS



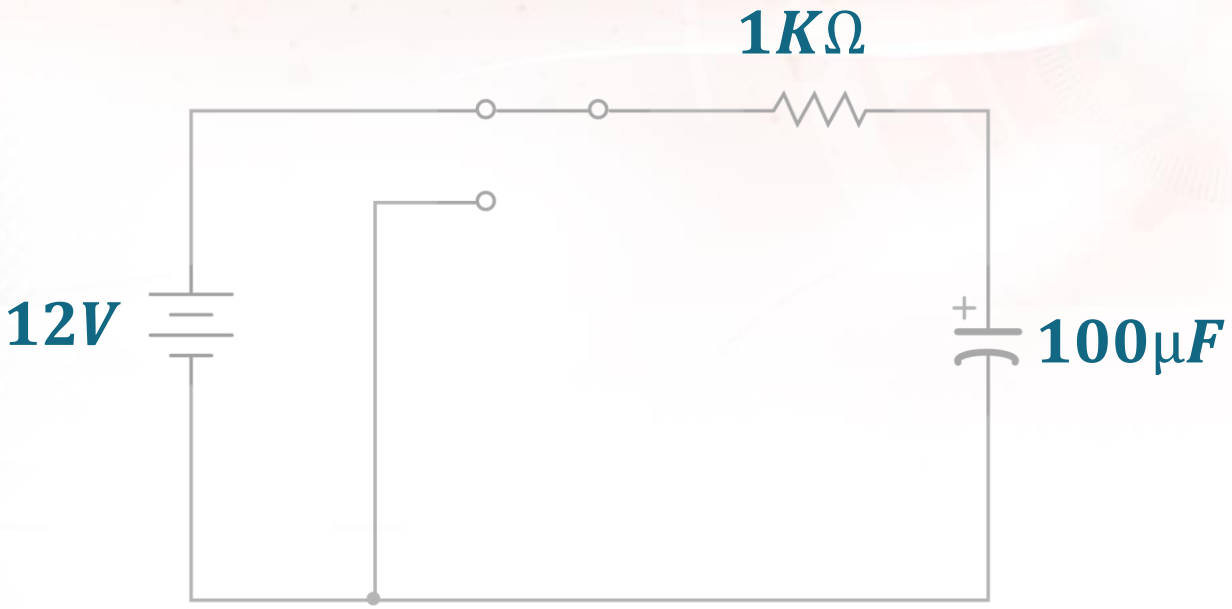
The voltage across the capacitor as a function of time,

$$v_c(t) = E(1 - e^{-t/\tau})$$

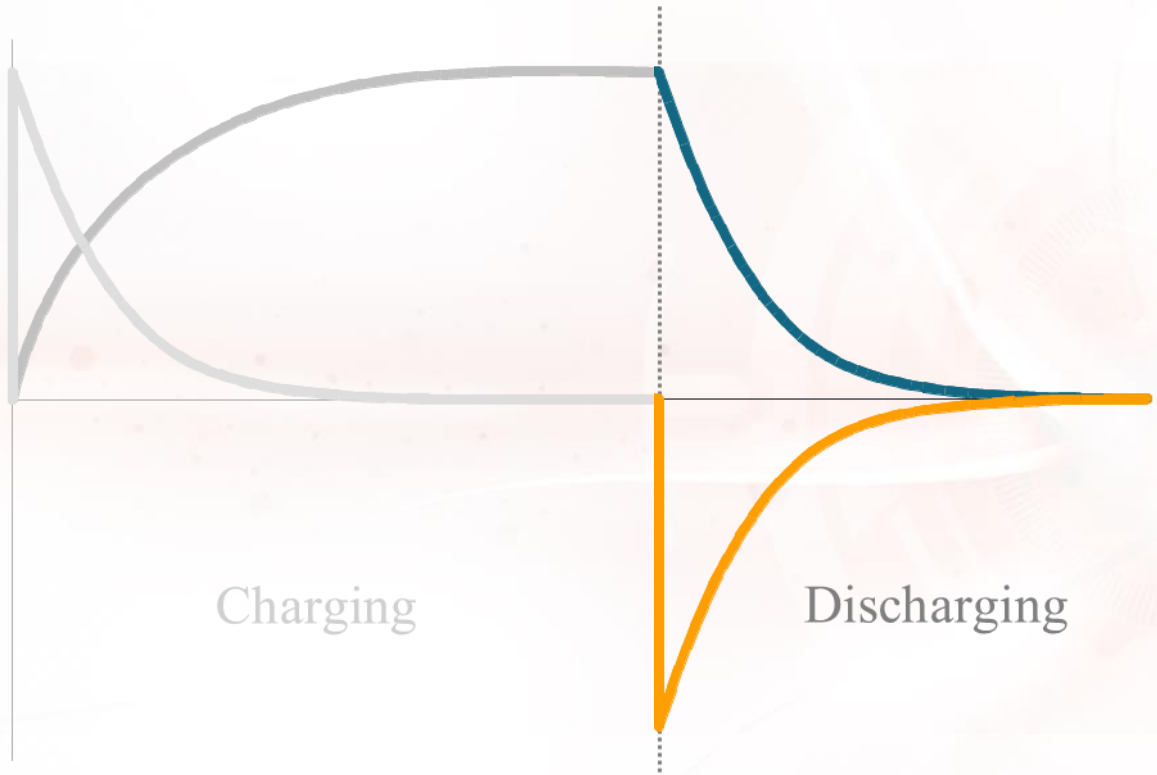
The current through the capacitor as a function of time,

$$i_c(t) = \frac{E}{R}\left(e^{-t/\tau}\right)$$

A **100uF** capacitor is connected to a **12V** DC power supply through a resistor of **1KΩ**. Determine the **voltage** across the capacitor after **200ms** of **charging**.



CAPACITOR DISCHARGING EQUATIONS



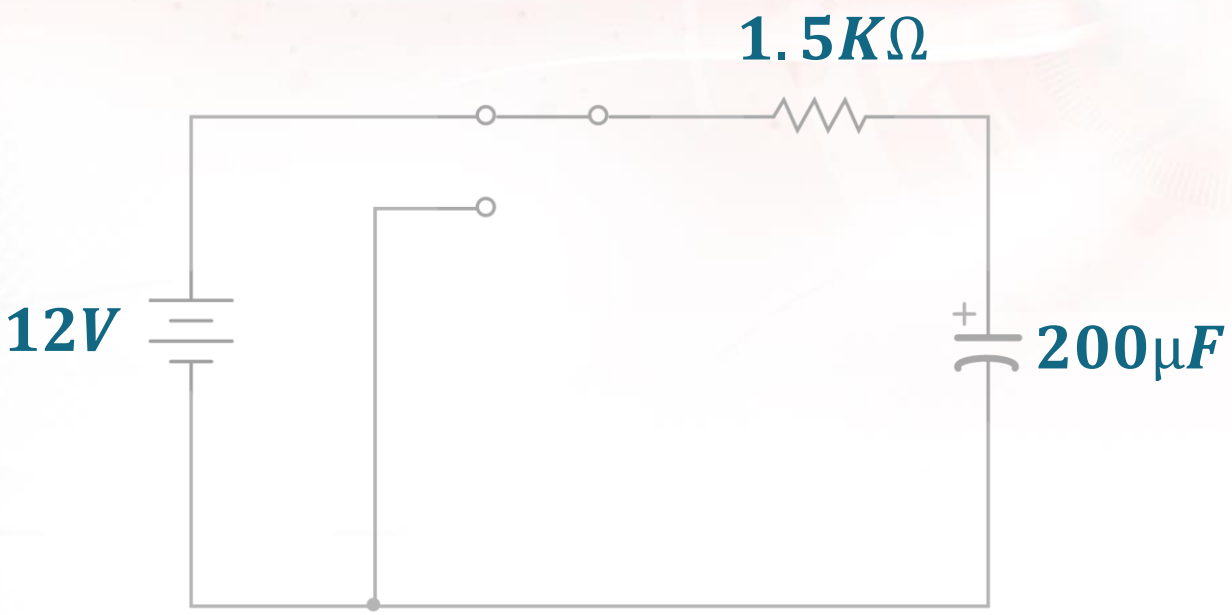
The voltage across the capacitor as a function of time

$$v_c(t) = Ee^{-t/\tau}$$

The current through the capacitor as a function of time

$$i_c(t) = -\frac{E}{R}\left(e^{-t/\tau}\right)$$

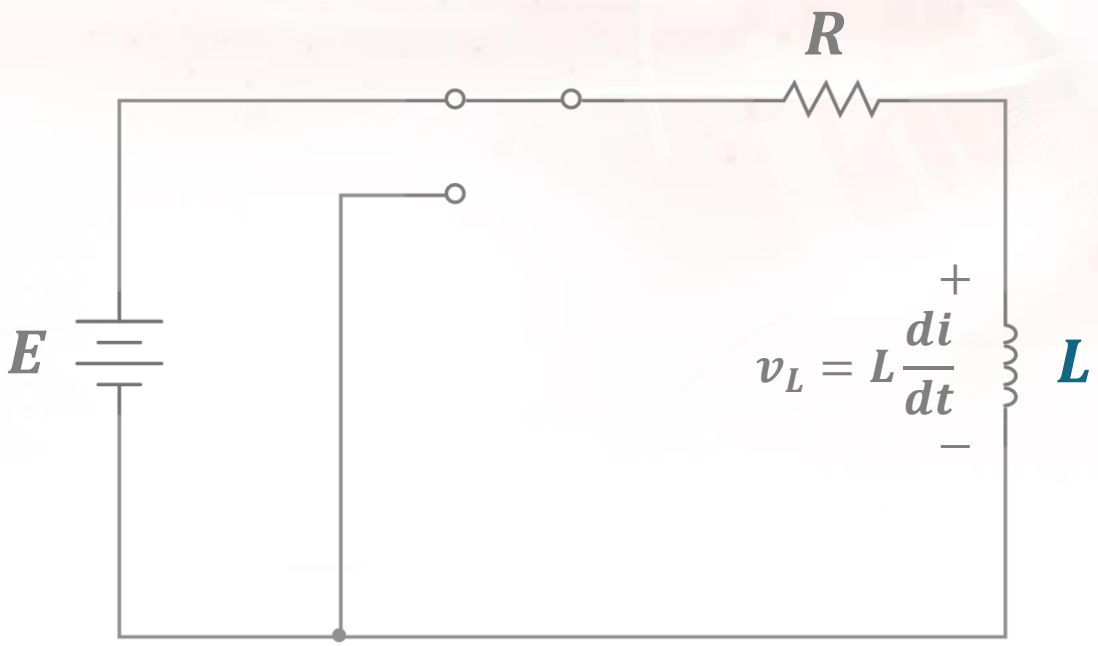
A **200μF** capacitor is initially charged to **12V**. It is then disconnected from the power supply and discharged a resistor of **1.5KΩ**. Determine the **voltage** across the capacitor after **0.1 seconds** of **discharging**.

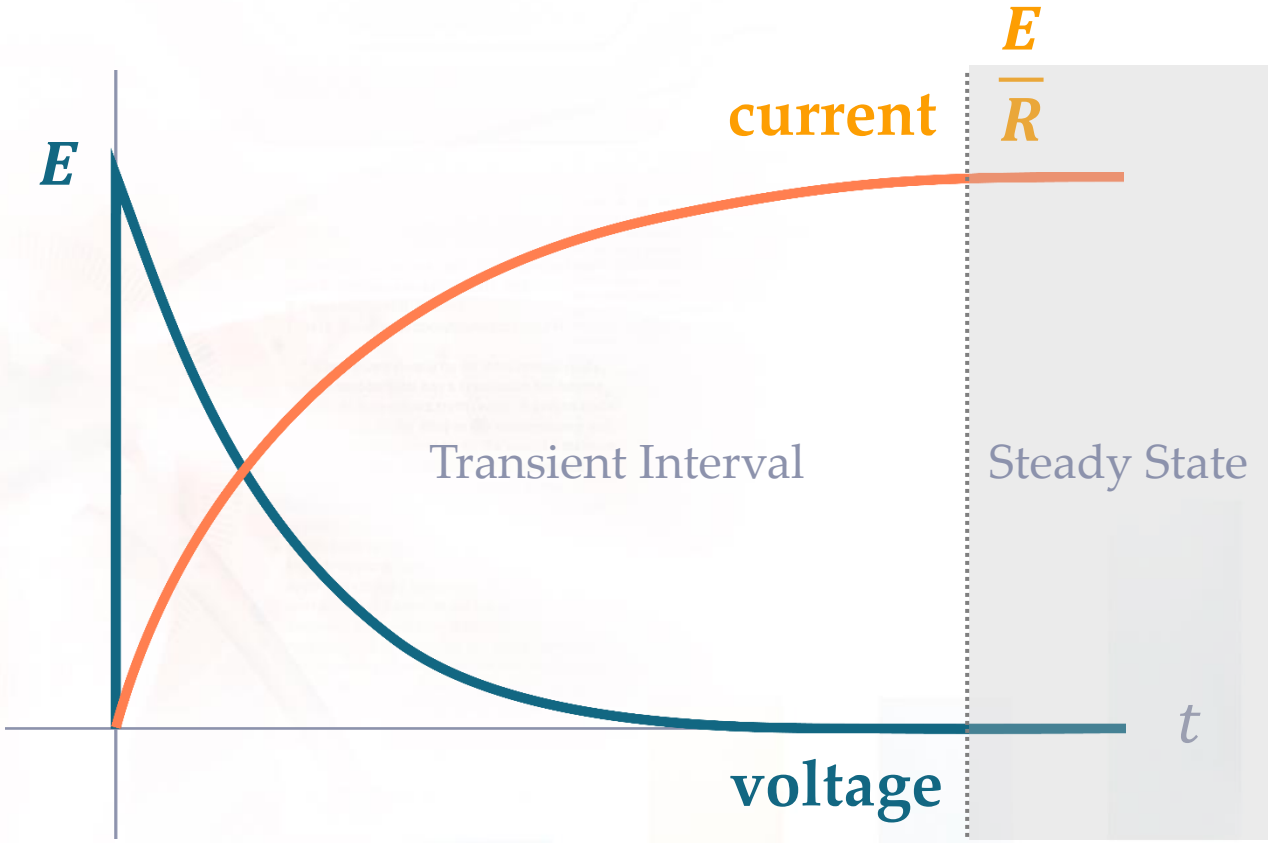


THE INDUCTOR TRANSIENT CURVE

Inductor **voltage** instantly jumps to E and exponentially decays to **zero**.

Inductor **current** begins at **zero** and exponentially increases to $\frac{E}{R}$ ampere.



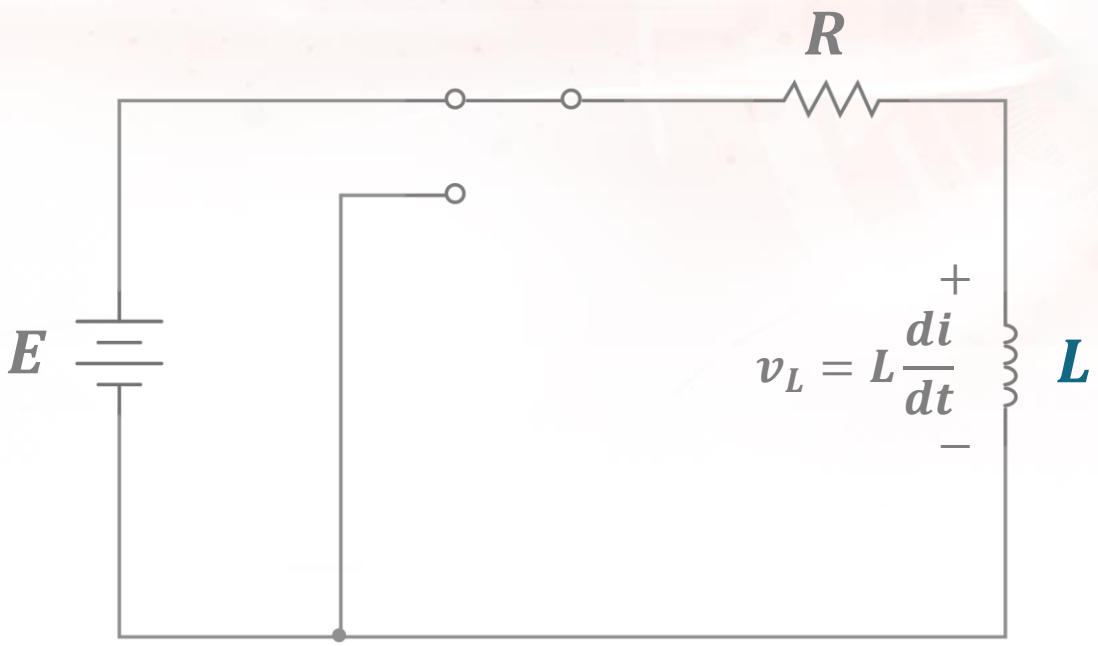


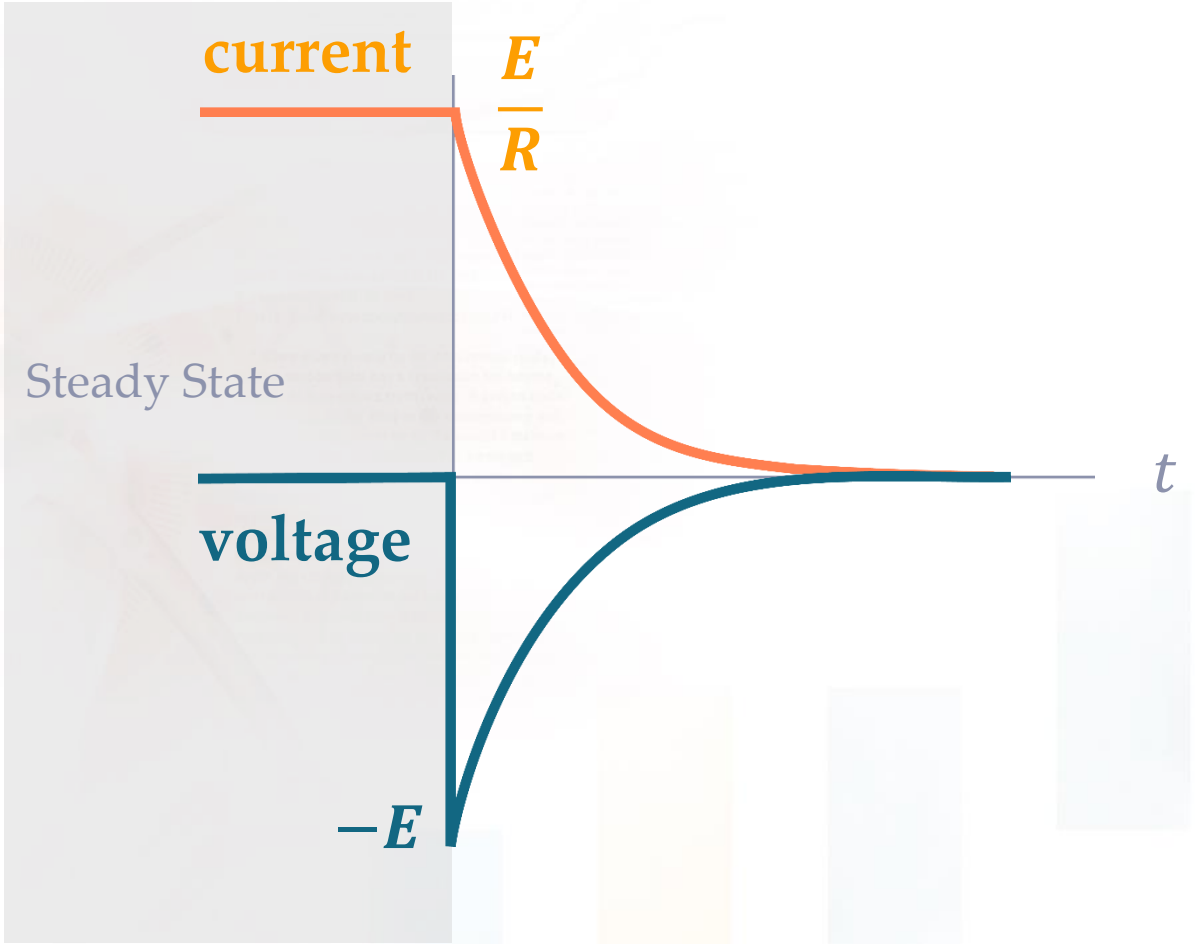
INDUCTOR DISCHARGING

Inductor **current** has $\frac{E}{R}$ ampere

Inductor **voltage** instantly jumps to $-E$

Both voltage and current will decay exponentially to **zero**

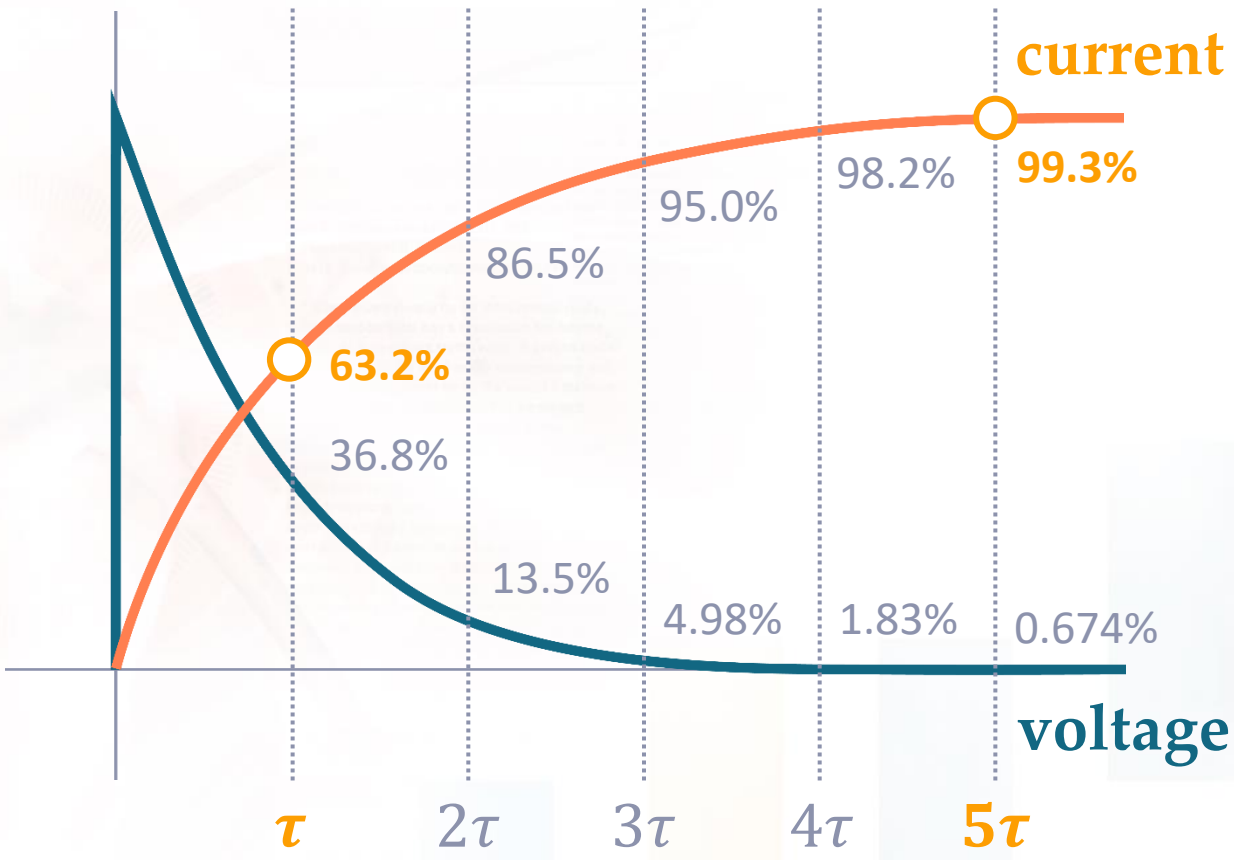




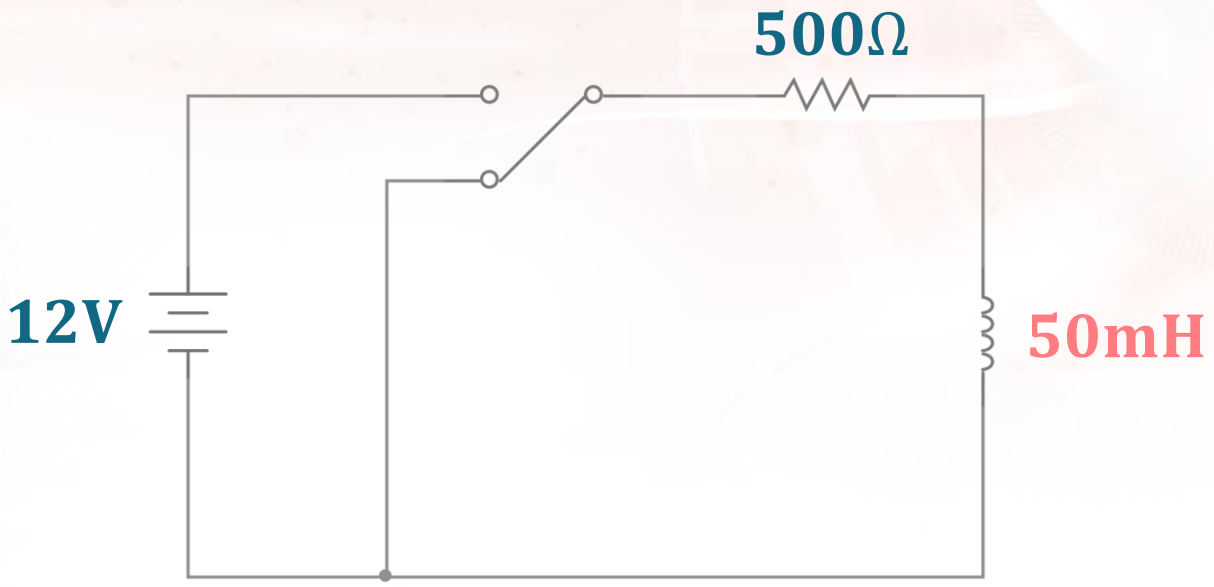
The rate at which capacitor charges depends on the ratio of L and R

$$\tau = \frac{L}{R}$$

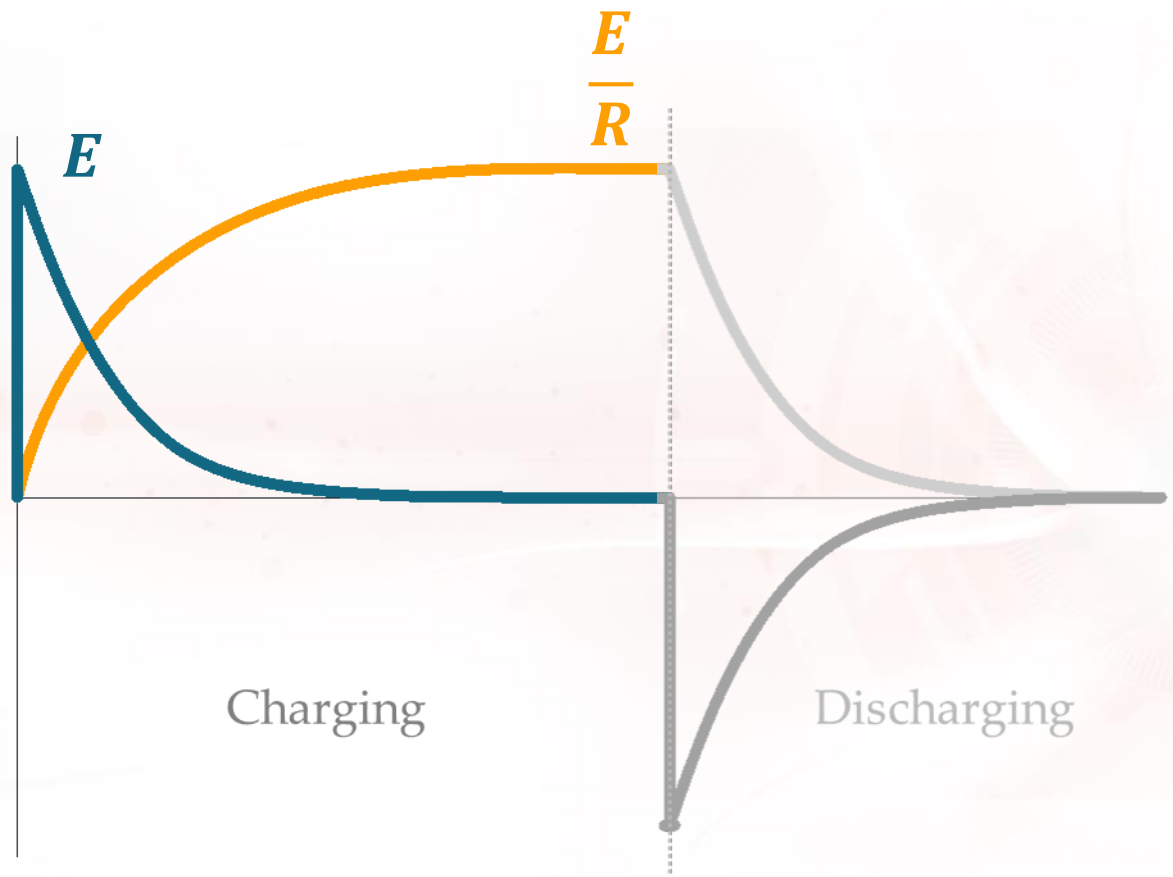
Transient can be considered to last for only five time constants, 5τ



A **50mH** inductor is connected to a **12V** DC power supply through a resistor of **500Ω**. Determine the **time** it takes for the inductor to charge to **95%** of its maximum current.



INDUCTOR CHARGING EQUATIONS



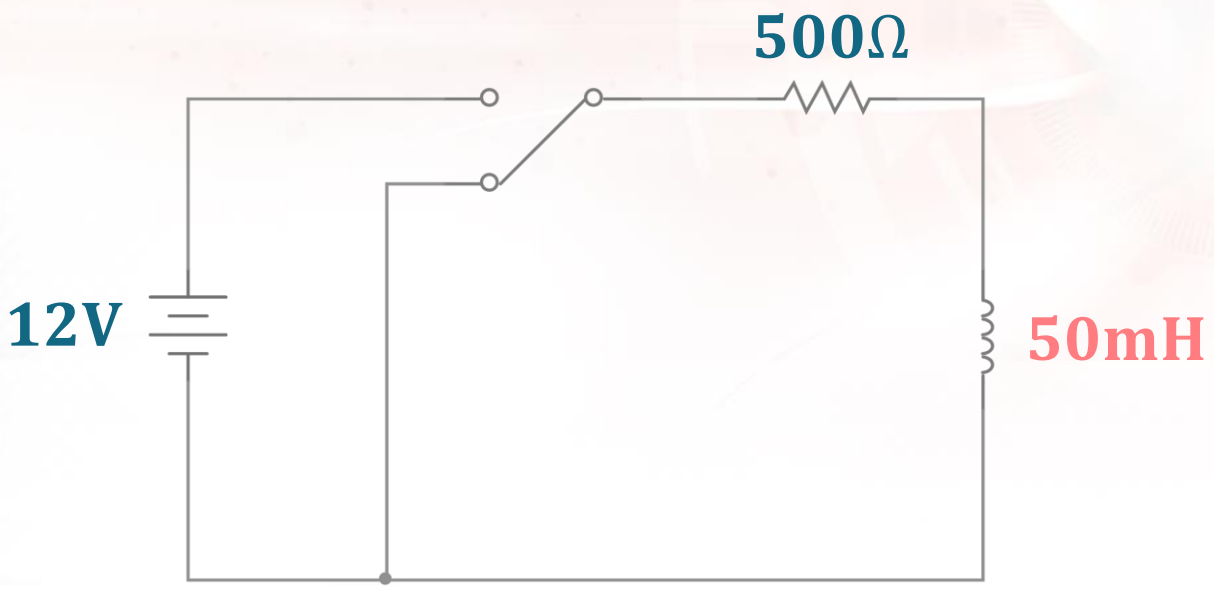
The current through the inductor as a function of time

$$i_L(t) = \frac{E}{R}(1 - e^{-t/\tau})$$

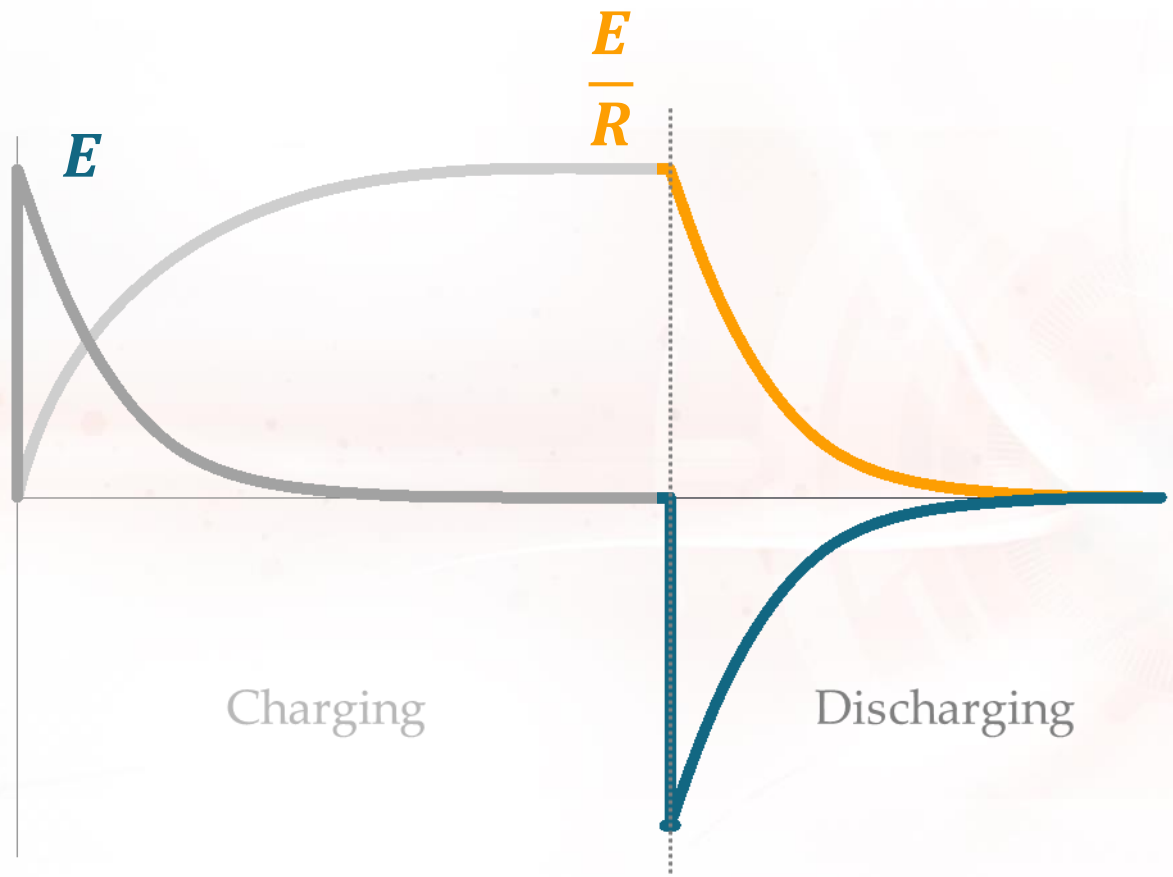
The voltage across the inductor as a function of time

$$v_L(t) = E\left(e^{-t/\tau}\right)$$

A **50mH** inductor is connected to a **12V** DC power supply through a resistor of **500Ω**. Determine the **current** through the inductor after **300μs** of **charging**.



INDUCTOR DISCHARGING EQUATIONS



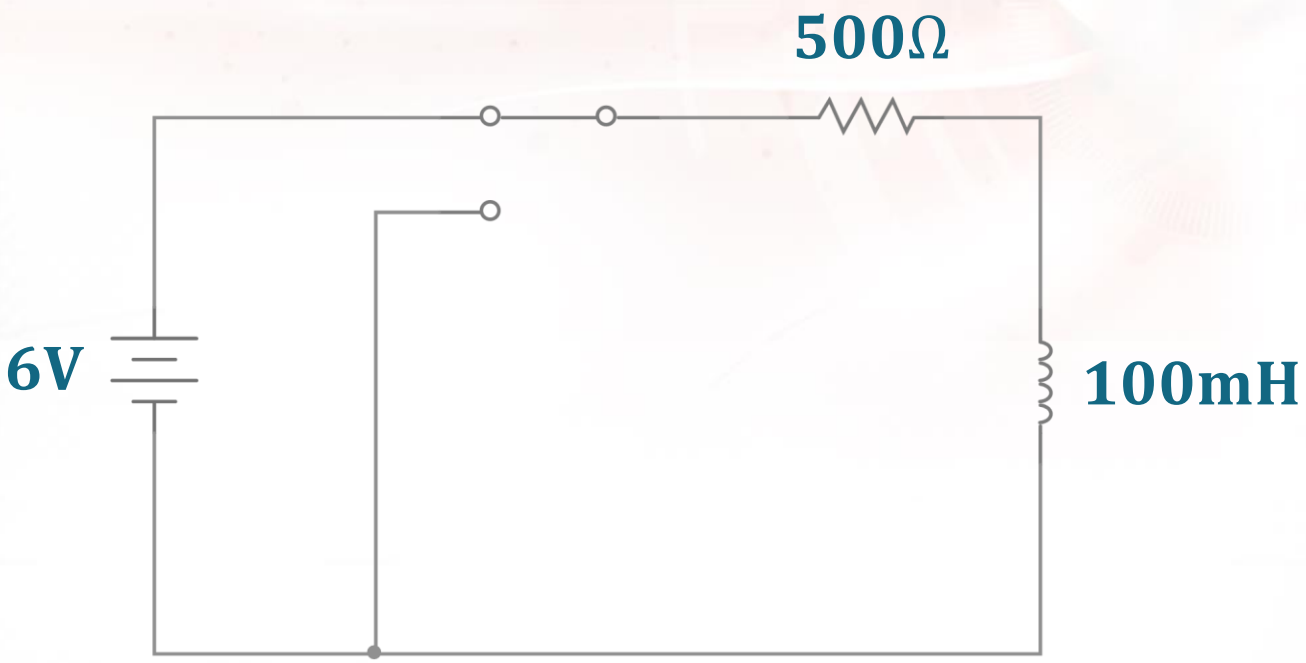
The current through the inductor as a function of time,

$$i_L(t) = \frac{E}{R} e^{-t/\tau}$$

The voltage across the inductor as a function of time

$$v_L(t) = -E\left(e^{-t/\tau}\right)$$

A **100mH** inductor is initially charged to **6V**. It is then disconnected from the power supply and discharged through a resistor of **500Ω**. Determine the **current** through the inductor after **0.02 seconds** of **discharging**.



LABORATORY